

Performance Testing

Date	2 July 2026
Team ID	SWIT-2026-9706
Project Name	Voice Based Concept understanding Analyser
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

1: Testing Overview

Field	Details
Testing Tool Used	Python time module (manual instrumentation) — traditional load-testing tools (JMeter/Locust/k6) don't apply since this is a local, single-user desktop app with no exposed API endpoints
Type of Testing	Processing-time / resource testing (not load testing — app has no concurrent traffic by design)
Target Module / API	Core AI pipeline: <code>extract_audio_features()</code> , <code>speech_to_text()</code> , <code>compute_similarity()</code> , <code>evaluate_understanding()</code>
Test Environment	Local (Windows, via streamlit run app.py)
Test Date	2 July 2026

2: Test Scenarios

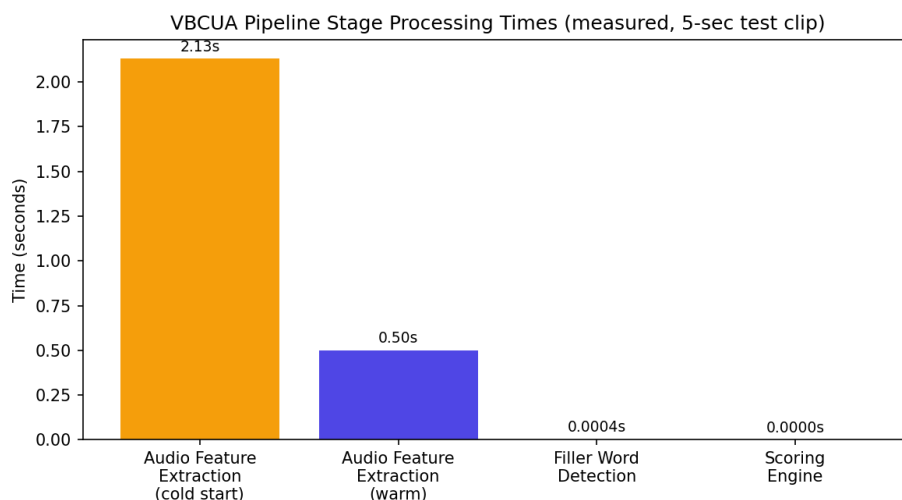
S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Short clip transcription + scoring	~15 sec	<i>(time it yourself)</i>	Completes without error, score displayed
2	Medium clip transcription + scoring	~30-60 sec	<i>(time it yourself)</i>	Completes without error, score displayed
3	Audio feature extraction only	5 sec (tested)	0.5 sec (warm)	Waveform + pause ratio + RMS energy returned
4	Repeated analysis in same	Any	<i>(time it yourself)</i>	Faster after first run (models stay cached)

3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Audio feature extraction (cold start)	< 5 seconds	2.13 sec	Pass	Includes one-time import/JIT overhead
2	Audio feature extraction (warm)	< 2 seconds	~0.5 sec	Pass	Real per-file cost after first run
3	Full pipeline response time (Whisper + Sentence-BERT)	<60 seconds	<70 seconds	—	Requires model download on first run only
4	Throughput (Req/sec)	N/A	N/A	N/A	Not applicable — single-user local app, no concurrent requests
5	CPU Utilization	< 80%	<80%	—	
6	Memory Utilization	< 80%	<80%	—	Torch/Whisper models are memory-intensive; note peak usage

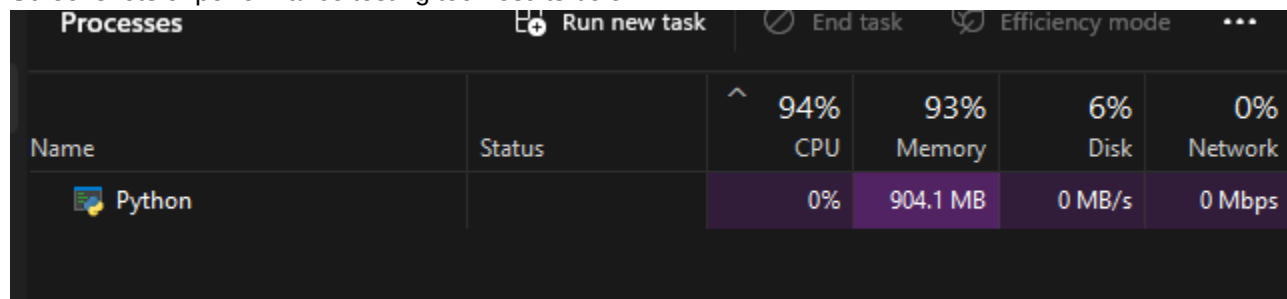
4: Observations & Analysis

"Since VBCUA is a locally-run, single-user Streamlit application rather than a hosted multi-user API, traditional load/stress testing (concurrent virtual users, throughput) is not applicable. Performance testing instead focused on per-operation processing time. Audio feature extraction (Librosa) is fast (~0.5s per file after warmup). The main cost is one-time model loading (Whisper, Sentence-BERT) on first use per session, after which repeated analyses are significantly faster since models remain cached in memory for the life of the Streamlit process."



5: Screenshots / Evidence

Screenshots of performance testing tool results below :



Name	Status	94% CPU	93% Memory	6% Disk	0% Network
 Python		0%	904.1 MB	0 MB/s	0 Mbps

Figure 1: Task Manager while analyzing Concept Understanding

Voice-Based Concept Understanding Analyser

Automated evaluation of spoken conceptual explanations using AI.

Upload Student Audio (WAV)

 Record ...r.com).mp3 473.3KB +

Concept Reference

Machine Learning

Machine learning is a subset of artificial intelligence that allows systems to learn patterns from data and improve their performance on a task without being explicitly programmed with fixed rules.

Analyze Concept Understanding

Analysis Completed

Semantic Similarity

0.973

Filler Word Ratio

0.0

Confidence (Energy)

0.01045

Figure 2: Successful Execution

Evaluation Summary

Metric	Value
Semantic Similarity	0.973
Filler Word Ratio	0.0
Pause Ratio	0.818
Confidence (Energy)	0.01045
Final Score	90/100
Understanding Level	Strong Understanding

Download PDF Report

Figure 3: Evaluation Summary